



Assignment of master's thesis

Title: Detection of Inflammatory Cells in Histological Images Using Computer Vision and Deep Learning Methods

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Study program: Informatics

Branch / specialization: Knowledge Engineering

Department: Department of Applied Mathematics

Validity: until the end of summer semester 2025/2026

Instructions

The detection of inflammatory cells in transplant organ histology images is a critical step in diagnosing transplant rejection and ensuring timely intervention. Traditional methods rely on manual examination by pathologists, which is labor-intensive, time-consuming, and prone to subjectivity. In recent years, Artificial Intelligence (AI) and Deep Learning (DL) techniques have shown great promise in automating image analysis, particularly in medical fields such as histopathology.

The objective of the thesis is the research and development of deep learning methods for histopathological image analysis, with a focus on cell detection or segmentation, and their classification for the deriving of inflammatory regions in transplant organ biopsies.

Analyze computer vision methods, particularly using deep neural networks (DNN), for detecting, segmenting, and classifying different cell types in histological images. Focus primarily on inflammatory cells in histological images from organ transplant biopsies. Conduct a literature review on the current state of research in this area. Propose and develop a solution for detecting inflammatory cells in histological images using DNN, also incorporating relevant interpretability methods. Implement the proposed solution using appropriate frameworks and libraries (e.g., PyTorch, OpenCV) and validate it on real data, such as publicly available datasets. Evaluate the solution using appropriate quantitative metrics and compare its performance with other existing methods.